

Question Bank

1. What are the important components of blockchain technologies? Discuss each component and their significance in detail.
2. What are digital signatures? Explain step by step process of signing and verification of documents using digital signatures.
3. Explain the process of simplified payment verification using neat sketch.
4. Draw neat sketch of block structure. Discuss each field and its significance in detail.
5. What is the relationship between difficulty target and nonce value? How is difficulty target value changed to maintain average block duration to 10 minutes in bitcoin?
6. What is markle root? How markle root value is calculated?
7. Draw neat sketch of bitcoin block structure. Discuss each field and its significance in detail.
8. Draw neat sketch of Ethereum block structure. Discuss each field and its significance in detail.
9. What are uncle blocks? What is their significance?
10. What is reward transaction? Why is it added to a block?
11. What is SHA-256?
12. What is mining? What is the need of mining in blockchain?
13. What is decentralized consensus? Why is it required in blockchain? How is decentralized consensus achieved in bitcoin?
14. What is the role of consensus algorithms in mining? Discuss various mining algorithms in bitcoin and Ethereum.
15. What is Proof of Work? Explain the process of implementing Proof of Work.
16. What is Proof of Stake? Explain the process of implementing Proof of Stake.
17. Differentiate between Proof of Work and Proof of Stake.
18. What is double spending attack?
19. How can double spending attack be prevented?
20. What is denial of service attack?
21. How can denial of service attack be prevented?
22. What are consensus attacks?
23. Explain various consensus attacks and prevention policies.
24. What is CAP Theorem? Prove CAP theorem using suitable example.
25. What are wallets in blockchain technology?
26. Discuss various types of wallets.
27. Differentiate between hot and cold wallets.
28. Give Blockchain applications with the help of case studies. Key Application Domains: Public/Government Sector, Healthcare Sector, Financial Sector, Supply Chain Management.
29. What are the future applications of blockchain technology? Discuss in detail.
30. What are the possible challenges to future applications of blockchain technology? Explain in detail.